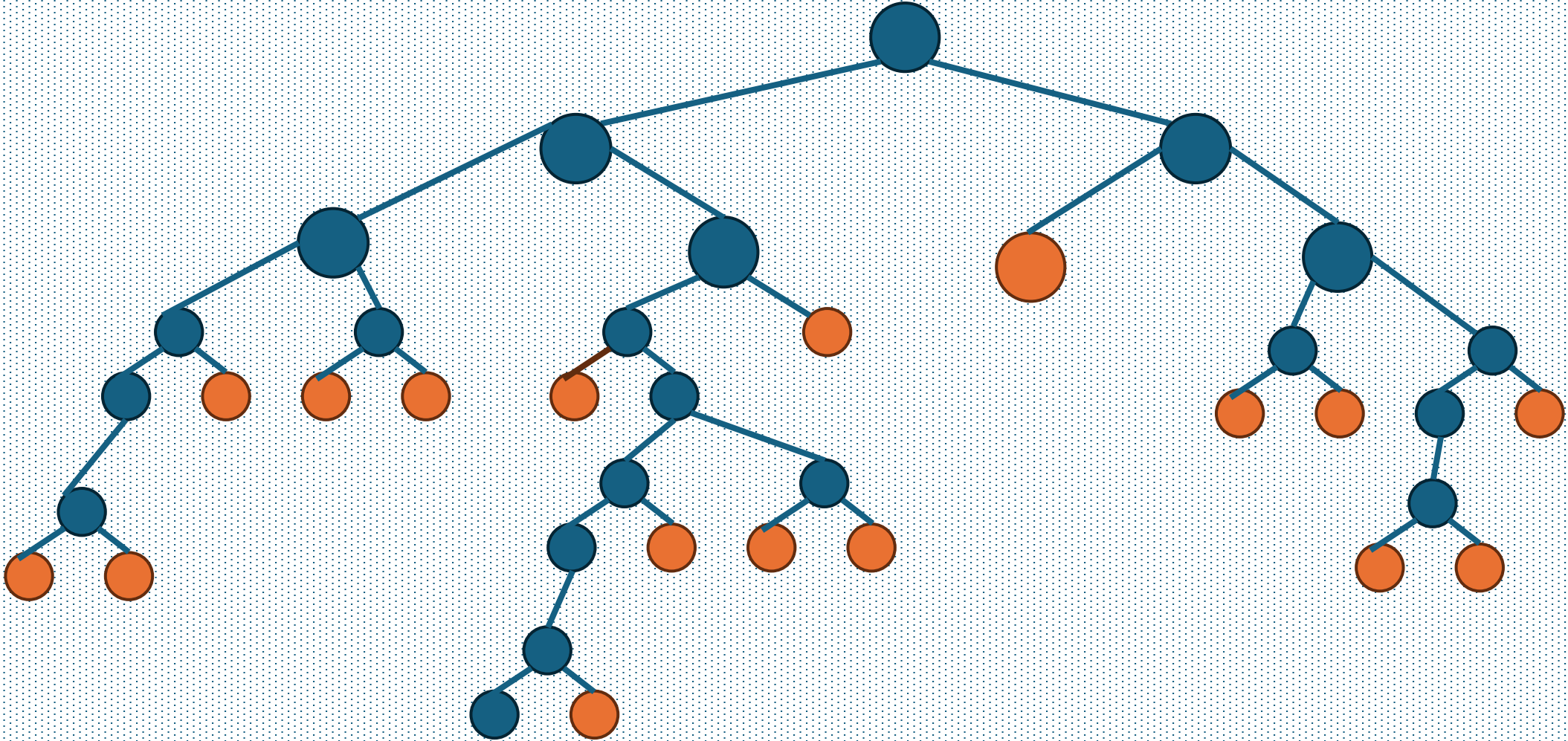


Decision TREE



A Decision Tree

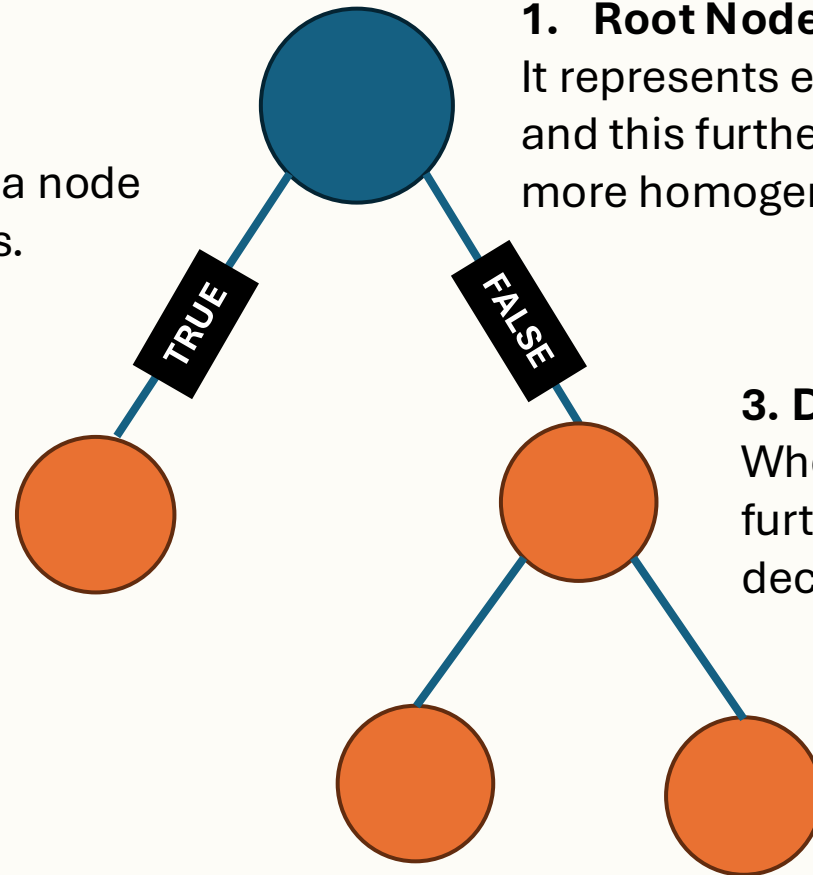
Terminology

2. Splitting

It is the process of dividing a node into two or more sub-nodes.

4. Leaf Node / Terminal Node

The node that does not split is called a Leaf Node.



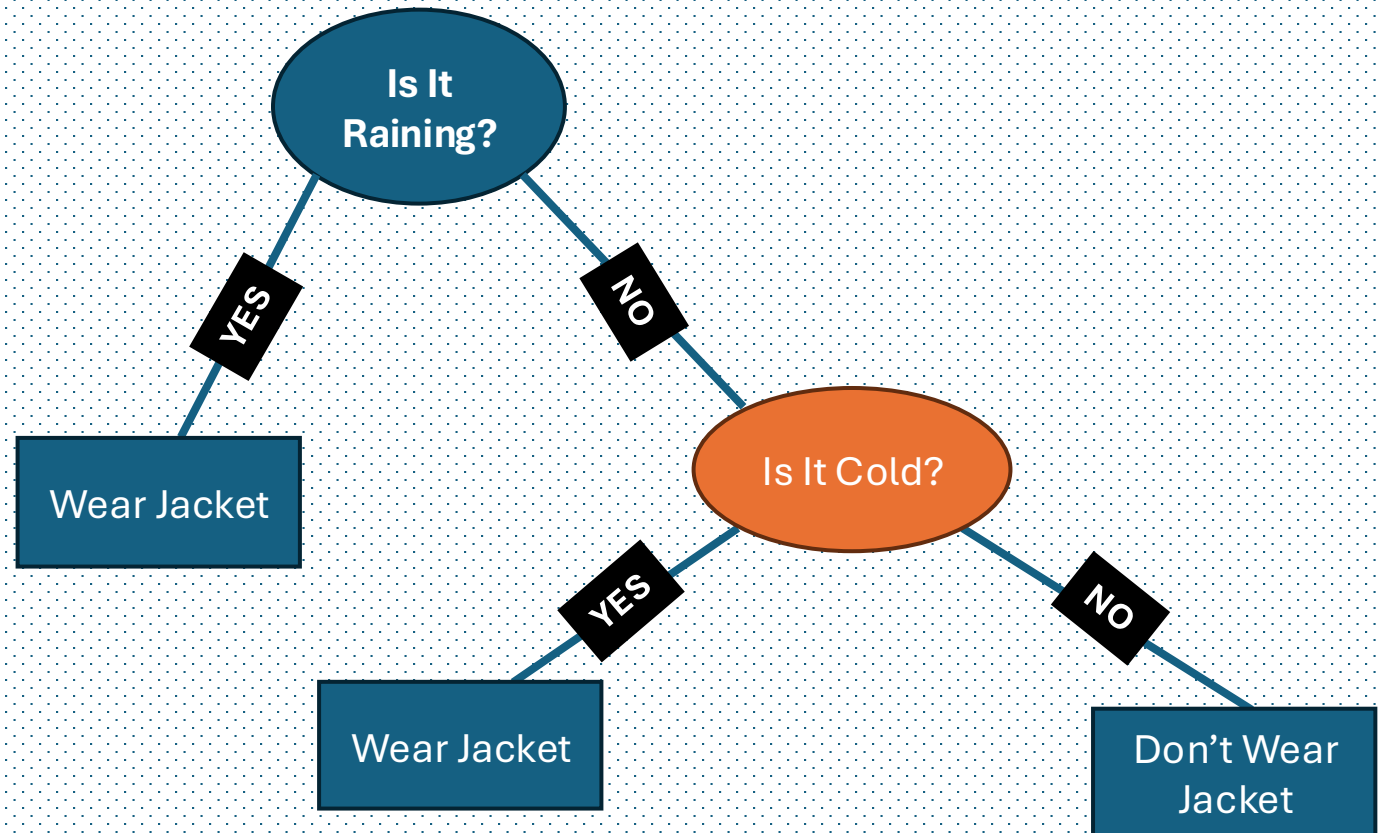
1. Root Node

It represents entire population or sample and this further gets divided into two or more homogenous sets.

3. Decision Node

When a sub-node splits into further sub-nodes, it is called a decision node.

Should I take a **Wear Jacket** 🧥 ?



CONSTRUCT DECISION TREE

1. Split Root Node

Select the decision rule which has the largest information gain to split the current node.

2. Split Child Node

Only select consider decision rule never selected before in the current branch

3. Repeat for each branch

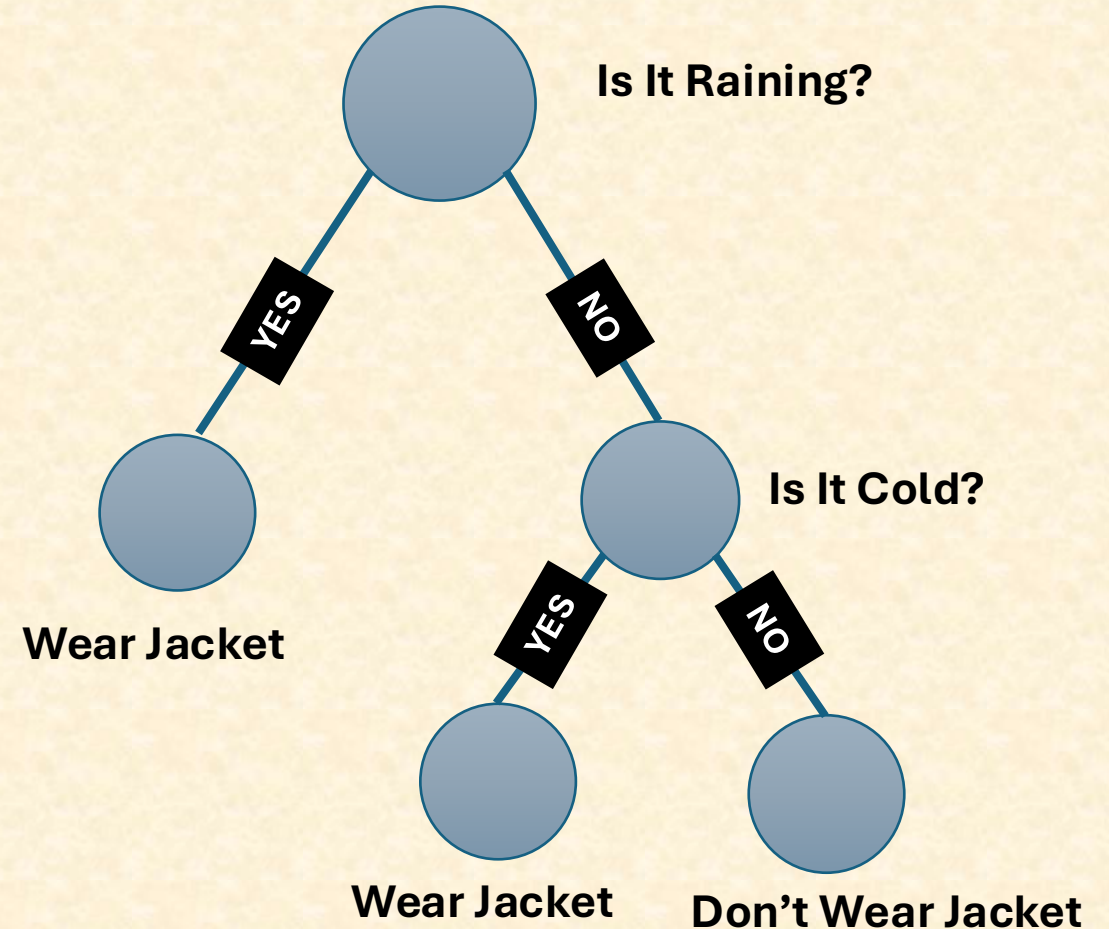
For each subset of the data:

- If it contains only one class: make it a **leaf node**.
- If more than one class: repeat the **splitting process**.

4. Stop Splitting

All samples belong to the same class.

No more features to split on.



Lets check following dataset.

Day	Humidity	Temp	Wind	Play
D1	High	Hot	Weak	No
D2	High	Hot	Strong	No
D3	High	Cool	Weak	Yes
D4	High	Cool	Strong	Yes
D5	Normal	Hot	Weak	Yes
D6	Normal	Cool	Weak	Yes
D7	Normal	Cool	Strong	No

Our task is to draw a decision tree .

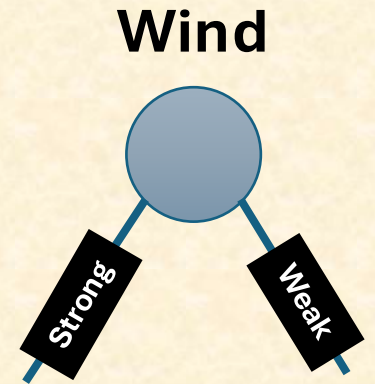
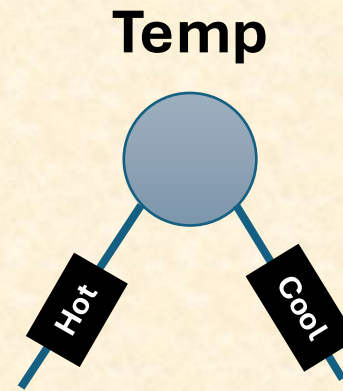
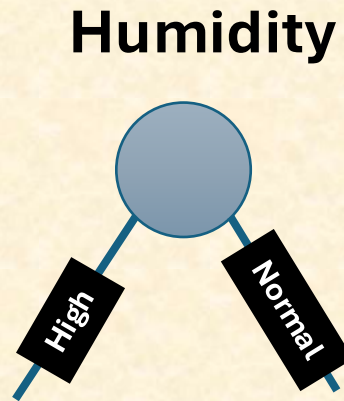
The Problem → is to find the root node. i.e. which condition is to check first.



How do we find the Root Node? 🤔

Solution:

1. Using **Information Gain**
2. Gini Index



Information Gain

Information Gain

- We want to determine **which attribute** in a given set of training feature vectors is **most useful** for discriminating between the classes to be learned
- **Information gain** tells **us how important a given attribute of the feature vectors is.**
- We will use it to decide the ordering of attributes in the nodes of a decision tree.

Information gain is given by:

$$Gain(T, X) = Entropy(T) - Entropy(T, X)$$

Where:

T = Target class

X = particular attribute

High information gain → the feature greatly reduces uncertainty and is useful for prediction.

Low information gain → the feature provides little information about the target.

Wait! But, What is **Entropy**?

Entropy is a measure of **impurity** or **uncertainty** in a dataset.

The entropy formula is:

$$H(S) = - \sum_{i=1}^n p_i \log_2(p_i)$$

Where:

$H(S)$ = entropy of dataset S

p_i = proportion of instances belonging to class i

n = number of classes

Entropy = 0 → dataset is perfectly pure (all samples belong to one class).

Higher entropy → classes are mixed, and uncertainty is higher.

For binary classification, the **maximum entropy is 1** when the classes are evenly split (50%-50%).

Example

Suppose a dataset contains:

8 Positive examples

8 Negative examples

Then:

$$p(\text{Positive}) = \frac{8}{16} = 0.5, \quad p(\text{Negative}) = \frac{8}{16} = 0.5$$

Entropy:

$$H(S) = -(0.5 \log_2 0.5 + 0.5 \log_2 0.5) = 1$$

This is the **maximum uncertainty**.

Now suppose a node contains:

16 Positive examples

0 Negative examples

Entropy:

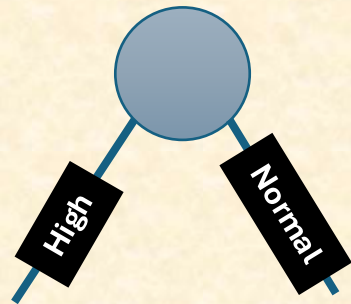
$$H(S) = -(1 \log_2 1) = 0$$

This node is **completely pure**.

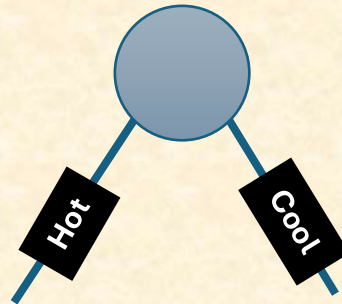
Let's go back to original problem

Day	Humidity	Temp	Wind	Play
D1	High	Hot	Weak	No
D2	High	Hot	Strong	No
D3	High	Cool	Weak	Yes
D4	High	Cool	Strong	Yes
D5	Normal	Hot	Weak	Yes
D6	Normal	Cool	Weak	Yes
D7	Normal	Cool	Strong	No

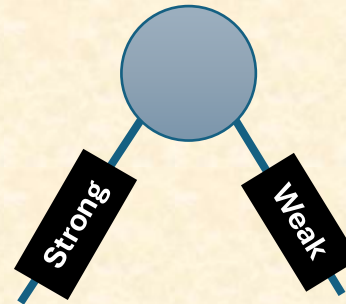
Humidity



Temp



Wind



Root Node?

Uncertainty? 🤔 🌟

Solution

We find the Information gain of each attribute like Temp, Humidity...

$$Gain(T, X) = Entropy(T) - Entropy(T, X)$$

Then we use the highest gain attribute as node.

$$Gain(Play, Humidity) = Entropy(Play) - Entropy(Play, Humidity)$$

$$\text{Gain}(\text{Play}, \text{Humidity}) = \text{Entropy}(\text{Play}) - \text{Entropy}(\text{Play}, \text{Humidity})$$

Dataset:

$$\log_2 = \log(x)/\log(2) = \log(x)/0.30103$$

Yes = 4

No = 3

$$\text{Entropy}(S) = - \sum p_i \log_2(p_i)$$

$$\text{Entropy}(\text{Play}) = -\frac{4}{7} \log_2 \frac{4}{7} - \frac{3}{7} \log_2 \frac{3}{7}$$

$$= -0.5714(-0.8074) - 0.4286(-1.2224)$$

$$= 0.4614 + 0.5239$$

$$\boxed{\text{Entropy}(\text{Play}) = 0.9852}$$

$$\text{Gain}(\text{Play}, \text{Humidity}) = \text{Entropy}(\text{Play}) - \text{Entropy}(\text{Play}, \text{Humidity})$$

$$\text{Entropy}(\text{Play}, \text{Humidity}) = P(\text{High}) * \text{Entropy}(\text{High}) + P(\text{Normal}) * \text{Entropy}(\text{Normal})$$

Entropy(High):

Dataset:

Yes = **2**

No = **2**

$$\begin{aligned} E(\text{High}) &= -\frac{2}{4} \log_2 \frac{2}{4} - \frac{2}{4} \log_2 \frac{2}{4} \\ &= 1 \end{aligned}$$

$$\text{Gain}(\text{Play}, \text{Humidity}) = \text{Entropy}(\text{Play}) - \text{Entropy}(\text{Play}, \text{Humidity})$$

$$\text{Entropy}(\text{Play}, \text{Humidity}) = P(\text{High}) * \text{Entropy}(\text{High}) + P(\text{Normal}) * \text{Entropy}(\text{Normal})$$

Entropy(Normal) :

Dataset:

Yes = **2**

No = **1**

$$E(\text{Normal}) = -\frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \frac{1}{3}$$

$$= 0.9183$$

$$\text{Gain}(\text{Play}, \text{Humidity}) = \text{Entropy}(\text{Play}) - \text{Entropy}(\text{Play}, \text{Humidity})$$

$$\text{Entropy}(\text{Play}, \text{Humidity}) = P(\text{High}) * \text{Entropy}(\text{High}) + P(\text{Normal}) * \text{Entropy}(\text{Normal})$$

Entropy(Play, Humidity):

Dataset:

High = 4

Normal = 3

$$E(\text{Play}, \text{Humidity}) = \frac{4}{7}(1) + \frac{3}{7}(0.9183)$$

$$= 0.5714 + 0.3936$$

$$= 0.9650$$

$$\textit{Gain}(\textit{Play}, \textit{Humidity}) = \textit{Entropy}(\textit{Play}) - \textit{Entropy}(\textit{Play}, \textit{Humidity})$$

Information Gain

$$\textit{IG}(\textit{Play}, \textit{Humidity}) = 0.9852 - 0.9650$$

$$\boxed{\textit{IG}(\textit{Play}, \textit{Humidity}) = 0.0202}$$

$$\text{Gain}(\text{Play}, \text{Temp}) = \text{Entropy}(\text{Play}) - \text{Entropy}(\text{Play}, \text{Temp})$$

$$\text{Entropy}(\text{Play}, \text{Temp}) = P(\text{Hot}) * \text{Entropy}(\text{Hot}) + P(\text{Cool}) * \text{Entropy}(\text{Cool})$$

Entropy(HOT):

Dataset:

Yes = **1**

No = **2**

$$\begin{aligned} E(\text{Hot}) &= -\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3} \\ &= 0.9183 \end{aligned}$$

$$\text{Gain}(\text{Play}, \text{Temp}) = \text{Entropy}(\text{Play}) - \text{Entropy}(\text{Play}, \text{Temp})$$

$$\text{Entropy}(\text{Play}, \text{Temp}) = P(\text{Hot}) * \text{Entropy}(\text{Hot}) + P(\text{Cool}) * \text{Entropy}(\text{Cool})$$

Entropy(Cool):

Dataset:

Yes = **3**

No = **1**

$$E(\text{Cool}) = -\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4}$$

$$= 0.8113$$

$$\text{Gain}(\text{Play}, \text{Temp}) = \text{Entropy}(\text{Play}) - \text{Entropy}(\text{Play}, \text{Temp})$$

$$\text{Entropy}(\text{Play}, \text{Temp}) = P(\text{Hot}) * \text{Entropy}(\text{Hot}) + P(\text{Cool}) * \text{Entropy}(\text{Cool})$$

Entropy(Play, Temp):

Dataset:

Hot = **3**

Cool = **4**

$$E(\text{Play}, \text{Temp}) = \frac{3}{7}(0.9183) + \frac{4}{7}(0.8113)$$

$$= 0.3936 + 0.4636$$

$$= 0.8571$$

$$Gain(Play, Temp) = Entropy(Play) - Entropy(Play, Temp)$$

Information Gain

$$IG(Temp) = 0.9852 - 0.8571$$

$$IG(Temp) = 0.1281$$

$$\text{Gain}(\text{Play}, \text{Wind}) = \text{Entropy}(\text{Play}) - \text{Entropy}(\text{Play}, \text{Wind})$$

$$\text{Entropy}(\text{Play}, \text{Wind}) = P(\text{Weak}) * \text{Entropy}(\text{Weak}) + P(\text{Strong}) * \text{Entropy}(\text{Strong})$$

Entropy(Weak):

Dataset:

Yes = **3**

No = **1**

$$E(\text{Weak}) = -\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4}$$

$$= 0.8113$$

$$\text{Gain}(\text{Play}, \text{Wind}) = \text{Entropy}(\text{Play}) - \text{Entropy}(\text{Play}, \text{Wind})$$

$$\text{Entropy}(\text{Play}, \text{Wind}) = P(\text{Weak}) * \text{Entropy}(\text{Weak}) + P(\text{Strong}) * \text{Entropy}(\text{Strong})$$

Entropy(Strong):

Dataset:

Yes = **1**

No = **2**

$$E(\text{Strong}) = -\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3}$$

$$= 0.9182$$

$$\text{Gain}(\text{Play}, \text{Wind}) = \text{Entropy}(\text{Play}) - \text{Entropy}(\text{Play}, \text{Wind})$$

$$\text{Entropy}(\text{Play}, \text{Wind}) = P(\text{Weak}) * \text{Entropy}(\text{Weak}) + P(\text{Strong}) * \text{Entropy}(\text{Strong})$$

Entropy(Play, Wind):

Dataset:

Weak = **4**

Strong = **3**

$$E(\text{Play}, \text{Wind}) = \frac{4}{7}(0.8113) + \frac{3}{7}(0.9182)$$

$$= 0.4636 + 0.3935$$

$$= 0.8571$$

$$Gain(Play, Wind) = Entropy(Play) - Entropy(Play, Wind)$$

Information Gain

$$IG(Wind) = 0.9852 - 0.8571$$

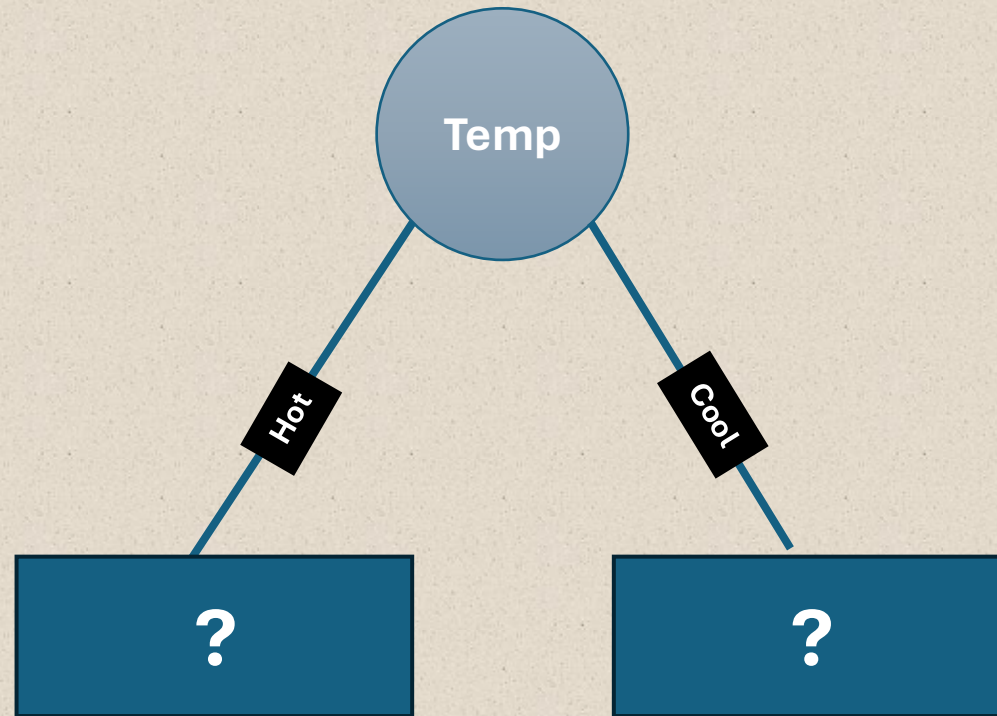
$$IG(Wind) = 0.1281$$

Compare Information Gain

Attribute	Information Gain
Temp	0.1281
Humidity	0.0202
Wind	0.1281

Temp and **Wind** are tied for the highest information gain (0.1281). We choose **Temp** as the root node.

So our root node is Temp.



Construct the decision tree for the target concept 'Play', using information gain.
Show each step and the final tree.

OR

For the given data, using the ID3 algorithm creates a decision tree for 'Play'.

OR

Construct the decision tree for the given dataset.

Now, Lets calculate Information gain for **Humidity** and **Wind** with respect to **Temp**
The data values are ***Hot*** and ***Cool***.

The new table with reference to *Hot* is:

Day	Humidity	Wind	Play
D1	High	Weak	No
D2	High	Strong	No
D5	Normal	Weak	Yes

Now next step is to calculate entropy and information gain.

$$Gain(Play_{hot} Humidity) = \textcolor{red}{Entropy(Play_{hot})} - Entropy(Play_{hot} Humidity)$$

Dataset:

$$\log_2 = \log(x)/\log(2) = \log(x)/0.30103$$

Yes = 1

No = 2

$$Entropy(S) = - \sum p_i \log_2(p_i)$$

$$Entropy(Play_{Hot}) = -\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3}$$

$$\textcolor{red}{Entropy(Play_{Hot})} = 0.918$$

$$\text{Gain}(\text{Play}_{\text{Hot}}, \text{Humidity}) = \text{Entropy}(\text{Play}_{\text{Hot}}) - \text{Entropy}(\text{Play}_{\text{Hot}}, \text{Humidity})$$

$$\text{Entropy}(\text{Play}_{\text{Hot}}, \text{Humidity}) = P(\text{High}) * \text{Entropy}(\text{High}) + P(\text{Normal}) * \text{Entropy}(\text{Normal})$$

Entropy(High):

Dataset:

Yes = 0

No = 2

$$\begin{aligned} E(\text{High}) &= -\frac{0}{2} \log_2 \frac{0}{2} - \frac{2}{2} \log_2 \frac{2}{2} \\ &= 0 \end{aligned}$$

$$\text{Gain}(\text{Play}_{\text{Hot}} \text{ Humidity}) = \text{Entropy}(\text{Play}_{\text{Hot}}) - \text{Entropy}(\text{Play}_{\text{Hot}} \text{ Humidity})$$

$$\text{Entropy}(\text{Play}_{\text{Hot}}, \text{Humidity}) = \text{P(High)} * \text{Entropy(High)} + \text{P(Normal)} * \text{Entropy(Normal)}$$

Entropy(Normal):

Dataset:

Yes = **1**

No = **0**

$$\begin{aligned} E(\text{Normal}) &= -\frac{1}{1} \log_2 \frac{1}{1} - \frac{0}{1} \log_2 \frac{0}{1} \\ &= 0 \end{aligned}$$

$$\text{Gain}(\text{Play}_{\text{Hot}} \text{ Humidity}) = \text{Entropy}(\text{Play}_{\text{Hot}}) - \text{Entropy}(\text{Play}_{\text{Hot}} \text{ Humidity})$$

$$\text{Entropy}(\text{Play}_{\text{Hot}}, \text{Humidity}) = P(\text{High}) * \text{Entropy}(\text{High}) + P(\text{Normal}) * \text{Entropy}(\text{Normal})$$

Dataset:

High = 2

Normal = 1

$$\text{Entropy}(\text{Play}_{\text{Hot}}, \text{Humidity}) = \frac{2}{3} 0 - \frac{1}{3} 0$$

$$= 0$$

$$\textit{Gain}(\textit{Play}_{\textit{Hot}} \textit{ Humidity}) = \textit{Entropy}(\textit{Play}_{\textit{Hot}}) - \textit{Entropy}(\textit{Play}_{\textit{Hot}} \textit{ Humidity})$$

Information Gain

$$\textit{IG}(\textit{Play}_{\textit{Hot}}, \textit{ Humidity}) = 0.918 - 0$$

$$\boxed{\textit{IG}(\textit{Play}_{\textit{Hot}}, \textit{ Humidity}) = 0.918}$$

$$Gain(Play_{Hot} \text{ Wind}) = Entropy(Play_{Hot}) - Entropy(Play_{Hot} \text{ Wind})$$

$$Entropy(Play_{Hot}, Wind) = P(\text{Weak}) * Entropy(\text{Weak}) + P(\text{Strong}) * Entropy(\text{Strong})$$

$Entropy(\text{Weak})$:

Dataset:

Yes = 1

No = 1

$$\begin{aligned} E(\text{Weak}) &= -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} \\ &= 1 \end{aligned}$$

$$Gain(Play_{Hot} \text{ Wind}) = Entropy(Play_{Hot}) - Entropy(Play_{Hot} \text{ Wind})$$

$$Entropy(Play_{Hot}, Wind) = P(\text{Weak}) * Entropy(\text{Weak}) + P(\text{Strong}) * Entropy(\text{Strong})$$

$Entropy(\text{Strong})$:

Dataset:

Yes = **0**

No = **1**

$$\begin{aligned} E(\text{Strong}) &= -\frac{0}{1} \log_2 \frac{0}{1} - \frac{1}{1} \log_2 \frac{1}{1} \\ &= 0 \end{aligned}$$

$$Gain(Play_{Hot} \text{ Wind}) = Entropy(Play_{Hot}) - Entropy(Play_{Hot} \text{ Wind})$$

$$Entropy(Play_{Hot}, Wind) = P(\text{Weak}) * Entropy(\text{Weak}) + P(\text{Strong}) * Entropy(\text{Strong})$$

Dataset:

Weak = 2

Strong = 1

$$Entropy(Play_{Hot}, Wind) = \frac{2}{3} 1 - \frac{1}{3} 0$$

$$= 0.667$$

$$Gain(Play_{Hot}, Wind) = Entropy(Play_{Hot}) - Entropy(Play_{Hot}, Wind)$$

Information Gain

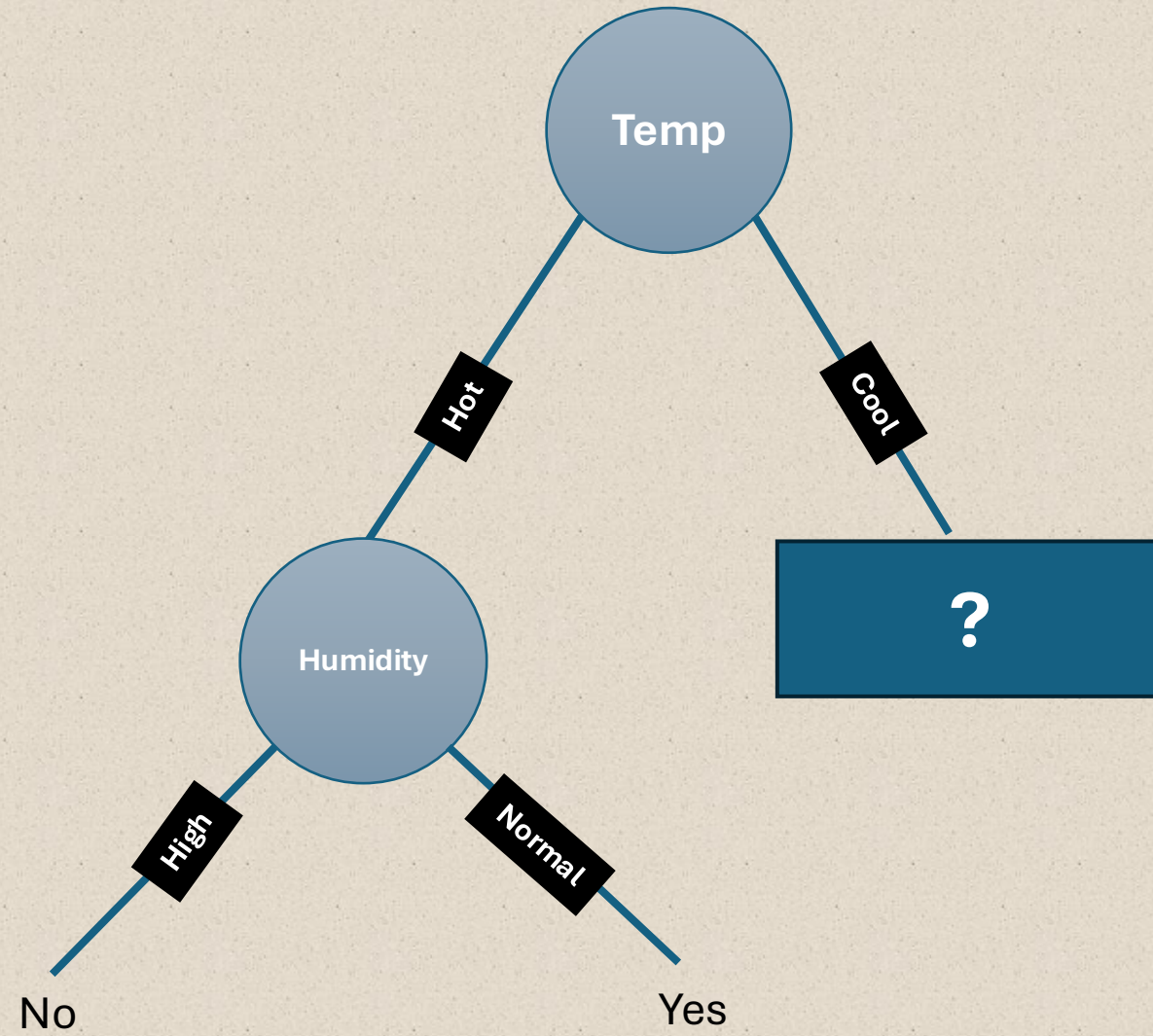
$$IG(Play_{Hot}, Wind) = 0.918 - 0.667$$

$$IG(Play_{Hot}, Wind) = 0.251$$

Compare Information Gain for Hot for Humidity and Wind

Attribute	Information Gain
Humidity	0.918
Wind	0.251

From the information gain we can clearly see for **Humidity** is **high** so we take it as parent node.



Similarly, we calculate Information gain for **Humidity** and **Wind** if **Temp** is **cool**

The new table with reference to *Cool* is:

Day	Humidity	Wind	Play
D3	High	Weak	Yes
D4	High	Strong	Yes
D6	Normal	Weak	Yes
D7	Normal	Strong	No

Now next step is to calculate entropy and information gain.

$$Gain(Play_{cool}, Humidity) = \textcolor{red}{Entropy(Play_{cool})} - Entropy(Play_{cool}, Humidity)$$

Dataset:

$$\log_2 = \log(x)/\log(2) = \log(x)/0.30103$$

Yes = 3

No = 1

$$Entropy(S) = - \sum p_i \log_2(p_i)$$

$$Entropy(Play_{cool}) = -\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4}$$

$$\textcolor{red}{Entropy(Play_{cool})} = 0.8112$$

$$\text{Gain}(\text{Play}_{\text{Cool}}, \text{Humidity}) = \text{Entropy}(\text{Play}_{\text{Cool}}) - \text{Entropy}(\text{Play}_{\text{Cool}}, \text{Humidity})$$

$$\text{Entropy}(\text{Play}_{\text{Cool}}, \text{Humidity}) = P(\text{High}) * \text{Entropy}(\text{High}) + P(\text{Normal}) * \text{Entropy}(\text{Normal})$$

Entropy(High):

Dataset:

Yes = **2**

No = **0**

$$\begin{aligned} E(\text{High}) &= -\frac{2}{2} \log_2 \frac{2}{2} - \frac{0}{2} \log_2 \frac{0}{2} \\ &= 0 \end{aligned}$$

$$Gain(Play_{Cool}, Humidity) = Entropy(Play_{Cool}) - Entropy(Play_{Cool}, Humidity)$$

$$Entropy(Play_{Cool}, Humidity) = P(High) * Entropy(High) + P(Normal) * Entropy(Normal)$$

$Entropy(Normal)$:

Dataset:

Yes = 1

No = 1

$$\begin{aligned} E(Normal) &= -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} \\ &= 1 \end{aligned}$$

$$\text{Gain}(\text{Play}_{\text{Cool}}, \text{Humidity}) = \text{Entropy}(\text{Play}_{\text{Cool}}) - \text{Entropy}(\text{Play}_{\text{Cool}}, \text{Humidity})$$

$$\text{Entropy}(\text{Play}_{\text{Cool}}, \text{Humidity}) = P(\text{High}) * \text{Entropy}(\text{High}) + P(\text{Normal}) * \text{Entropy}(\text{Normal})$$

Dataset:

High = 2

Normal = 2

$$\text{Entropy}(\text{Play}_{\text{Cool}}, \text{Humidity}) = \frac{2}{4} 0 - \frac{2}{4} 1$$

$$= 0.5$$

$$\textit{Gain}(\textit{Play}_{\textit{Cool}}, \textit{Humidity}) = \textit{Entropy}(\textit{Play}_{\textit{Cool}}) - \textit{Entropy}(\textit{Play}_{\textit{Cool}}, \textit{Humidity})$$

Information Gain

$$\textit{IG}(\textit{Play}_{\textit{Cool}}, \textit{Humidity}) = 0.8112 - 0.5$$

$$\boxed{\textit{IG}(\textit{Play}_{\textit{Cool}}, \textit{Humidity}) = 0.3112}$$

$$\text{Gain}(\text{Play}_{\text{Cool}}, \text{Wind}) = \text{Entropy}(\text{Play}_{\text{Cool}}) - \text{Entropy}(\text{Play}_{\text{Cool}}, \text{Wind})$$

$$\text{Entropy}(\text{Play}_{\text{Cool}}, \text{Wind}) = P(\text{Weak}) * \text{Entropy}(\text{Weak}) + P(\text{Strong}) * \text{Entropy}(\text{Strong})$$

Entropy(Weak):

Dataset:

Yes = **2**

No = **0**

$$\begin{aligned} E(\text{Weak}) &= -\frac{2}{2} \log_2 \frac{2}{2} - \frac{0}{2} \log_2 \frac{0}{2} \\ &= 0 \end{aligned}$$

$$\text{Gain}(\text{Play}_{\text{Cool}}, \text{Wind}) = \text{Entropy}(\text{Play}_{\text{Cool}}) - \text{Entropy}(\text{Play}_{\text{Cool}}, \text{Wind})$$

$$\text{Entropy}(\text{Play}_{\text{Cool}}, \text{Wind}) = P(\text{Weak}) * \text{Entropy}(\text{Weak}) + P(\text{Strong}) * \text{Entropy}(\text{Strong})$$

$\text{Entropy}(\text{Strong})$:

Dataset:

Yes = 1

No = 1

$$\begin{aligned} E(\text{Strong}) &= -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} \\ &= 1 \end{aligned}$$

$$\text{Gain}(\text{Play}_{\text{Cool}}, \text{Wind}) = \text{Entropy}(\text{Play}_{\text{Cool}}) - \text{Entropy}(\text{Play}_{\text{Cool}}, \text{Wind})$$

$$\text{Entropy}(\text{Play}_{\text{Cool}}, \text{Wind}) = P(\text{Weak}) * \text{Entropy}(\text{Weak}) + P(\text{Strong}) * \text{Entropy}(\text{Strong})$$

Dataset:

Weak = 2

Strong = 2

$$\text{Entropy}(\text{Play}_{\text{Cool}}, \text{Wind}) = \frac{2}{4} 0 - \frac{2}{4} 1$$

$$= 0.5$$

$$Gain(Play_{Cool}, Wind) = Entropy(Play_{Cool}) - Entropy(Play_{Cool}, Wind)$$

Information Gain

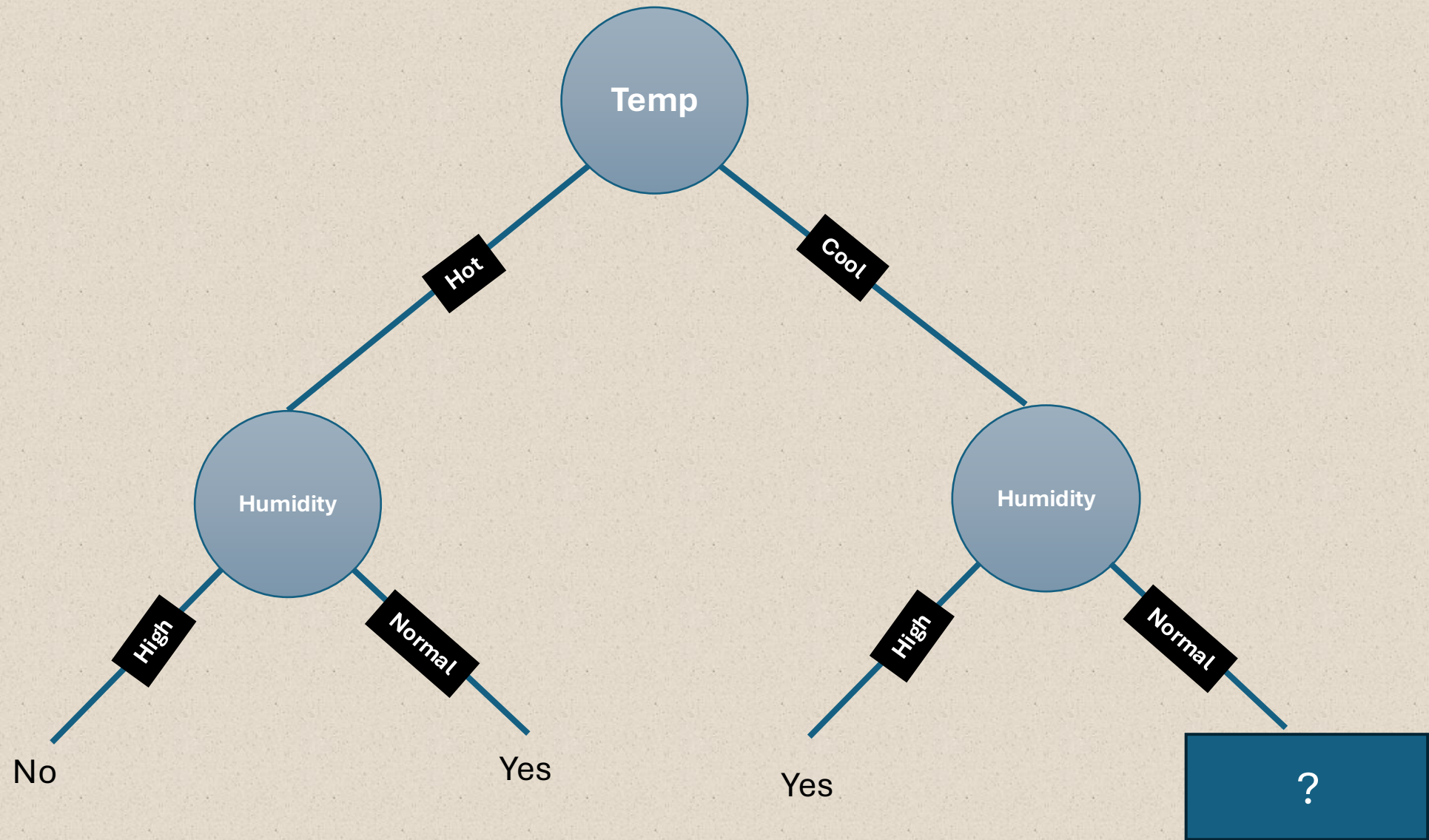
$$IG(Play_{Cool}, Wind) = 0.8112 - 0.5$$

$$IG(Play_{Cool}, Wind) = 0.3112$$

Compare Information Gain for Hot for Humidity and Wind

Attribute	Information Gain
Humidity	0.3112
Wind	0.3112

Since both humidity and wind has same information gain, we choose **humidity** as parent node.



Similarly, we calculate Information gain for **Temp = cool** and **Humidity = Normal**

Day	Wind	Play
D6	Weak	Yes
D7	Strong	No

Since only left attribute is we can draw node directly

Final Decision Tree

